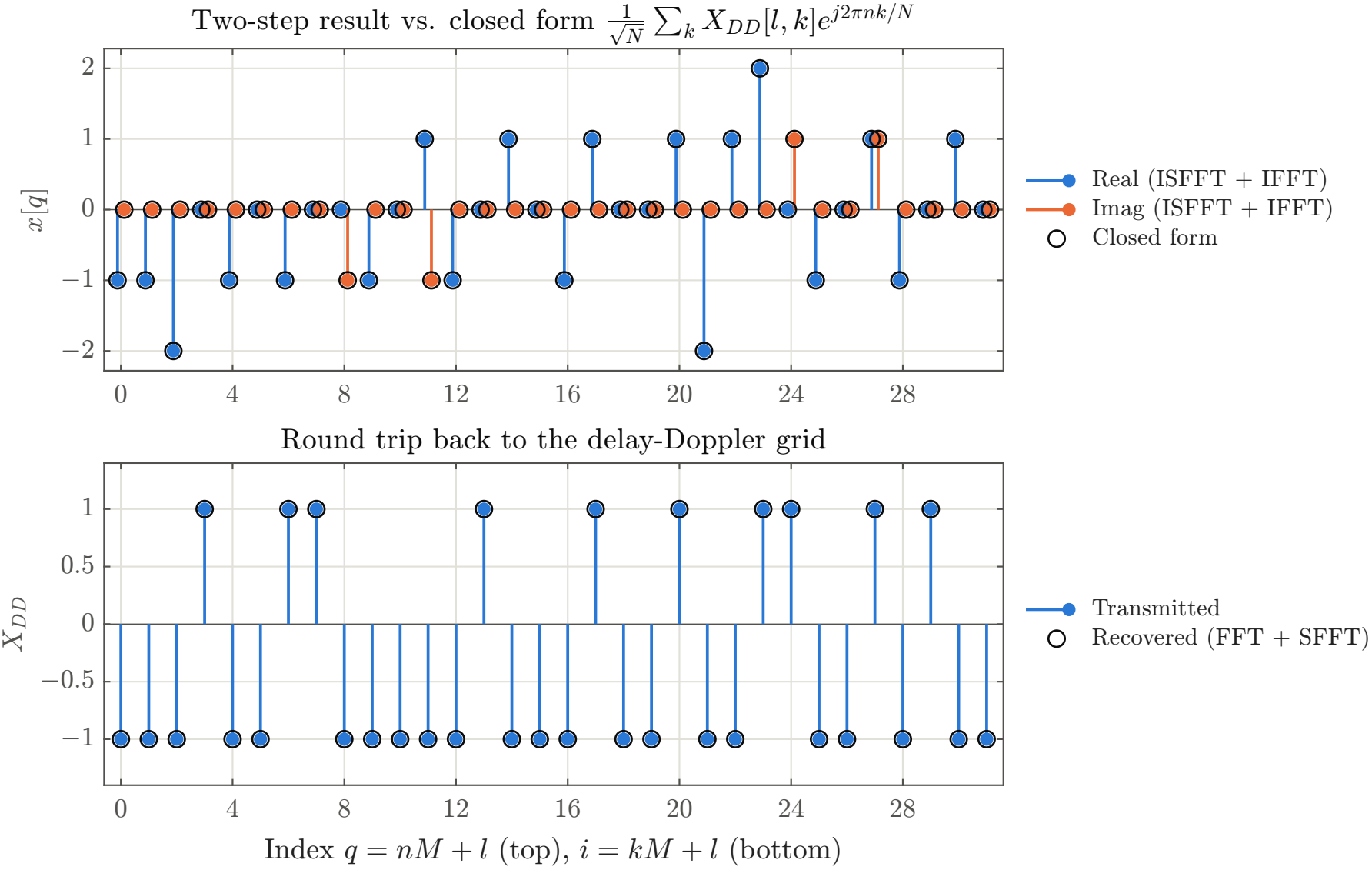
Two-step result vs. closed form $\frac{1}{\sqrt{N}} \sum_k X_{DD}[l,k] e^{j2\pi nk/N}$
$x[q]$
2
1
0
−1
−2
0
4
8
12
16
20
24
28
Real (ISFFT + IFFT)
Imag (ISFFT + IFFT)
Closed form
Round trip back to the delay-Doppler grid
X_{DD}
1
0.5
0
−0.5
−1
0
4
8
12
16
20
24
28
Transmitted
Recovered (FFT + SFFT)
Index $q = nM + l$ (top), $i = kM + l$ (bottom)